



Channel Model for mmWave (5G)



1. Definition

- A **mmWave channel model** is a **mathematical representation** of how high-frequency (mmWave) signals propagate in real environments.
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Simple Meaning

👉 Instead of testing signals everywhere:

- Engineers create a **model (simulation)**
 - To understand how mmWave behaves
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◆ 2. Why Special Channel Model Needed for mmWave?

👉 Because mmWave is **different from normal signals**



Reasons

- Very high frequency (30–300 GHz)
 - Short range
 - Easily blocked
 - Sensitive to environment
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👉 So normal models don't work ❌

👉 Need **special mmWave models** ✔️

◆ 3. Characteristics of mmWave Channel

Key Points

◆ High Path Loss

- Signal reduces quickly

◆ Directional Transmission

- Uses beamforming

◆ Sparse Multipath

- Few strong paths (not many reflections)

◆ Blockage Sensitive

- Easily blocked by objects

◆ Atmospheric Absorption

- Rain, oxygen absorb signals
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◆ 4. Types of mmWave Channel Models

◆ 1. Deterministic Model

✓ Meaning

- Based on **actual environment (buildings, roads)**

📌 Features

- Very accurate
 - Uses ray tracing
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◆ 2. Stochastic Model

✓ Meaning

- Based on **probability and statistics**

📌 Features

- Less complex
 - Used for large networks
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◆ 3. Geometry-Based Stochastic Model (GBSM)** ★ Important

✓ Meaning

- Combines geometry + probability

📌 Features

- Most widely used in 5G
 - Realistic and flexible
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◆ 5. Basic mmWave Channel Model Representation

👉 Channel includes:

- Line-of-Sight (LOS) path
 - Non-Line-of-Sight (NLOS) paths
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Diagram

 → → →  (LOS)
↘ → → →
↘ → (NLOS paths)

◆ 6. Key Components in mmWave Channel Model

Important Elements

- Path loss model
 - Delay spread
 - Angular spread
 - Doppler shift
 - Blockage model
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◆ 7. Challenges in mmWave Channel Modelling

Points

- High path loss modelling
 - Blockage modelling
 - Mobility handling
 - Accurate environment representation
 - Complexity
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8. Applications of mmWave Channel Model

- 5G network design
 - Performance analysis
 - Coverage planning
 - Simulation studies
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9. Importance

- Helps improve signal quality
 - Reduces network failures
 - Enables efficient design
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